

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$1 + C_1 y = \sqrt[3]{\dots}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$9) \cos y = x - 3$$

$$y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \ln(2x^2 - 3), x > \frac{\sqrt{6}}{2}$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x < 0$$

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$1 + C_1 y = \sqrt[3]{\dots}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$9) \cos y = x - 3$$

$$y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \ln(2x^2 - 3), x > \frac{\sqrt{6}}{2}$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x < 0$$

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$1 + C_1 y = \sqrt[3]{\dots}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$9) \cos y = x - 3$$

$$y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \ln(2x^2 - 3), x > \frac{\sqrt{6}}{2}$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x < 0$$

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \sqrt[3]{e^3 + 3e^2}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$1 + C_1 y = \sqrt[3]{\dots}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$9) \cos y = x - 3$$

$$y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \ln(2x^2 - 3), x > \frac{\sqrt{6}}{2}$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x < 0$$



Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$1 + C_1 y = \sqrt[3]{x}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$9) \cos y = x - 3$$

$$y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \ln(2x^2 - 3), x > \frac{\sqrt{6}}{2}$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x < 0$$

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$1 + C_1 y = \sqrt[3]{\dots}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$9) \cos y = x - 3$$

$$y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \ln(2x^2 - 3), x > \frac{\sqrt{6}}{2}$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x < 0$$

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$\frac{y^3}{3} = \frac{x^4}{4} + C$$

$$= \sqrt[3]{\frac{3x^4}{4}} + C$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$5) x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$9) y = x - 3 \quad y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$y = \frac{4\ln(2x^3 - 1)}{2}, x > \frac{\sqrt[3]{6}}{2}$$

$$12) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$y = \frac{\ln(2x^3 - 1)}{2}, x > \sqrt[3]{\frac{1}{2}}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x < 0$$

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \sqrt[3]{e^3 + 3e^2}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$4 - + C_1 y = \sqrt[3]{}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$: x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$5) y = x - 3 \quad y = \cos^{-1}(x - 3), 2 <$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \frac{4 \ln(2x^2 - 3)}{2}, x > \frac{\sqrt{6}}{2}$$

$$12) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x <$$



Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \sqrt[3]{e^3 + 3e^2}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$4 - + C_1 y = \sqrt[3]{}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$: x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$5) y = x - 3 \quad y = \cos^{-1}(x - 3), 2 <$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \frac{4 \ln(2x^2 - 3)}{2}, x > \frac{\sqrt{6}}{2}$$

$$12) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x <$$

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \sqrt[3]{e^3 + 3e^2}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$- + C_1 y = \sqrt[3]{}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$: x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$5) y = x - 3 \quad y = \cos^{-1}(x - 3), 2 <$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \frac{4 \ln(2x^2 - 3)}{2}, x > \frac{\sqrt{6}}{2}$$

$$12) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x <$$

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \sqrt[3]{e^3 + 3e^2}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$- + C_1 y = \sqrt[3]{\phantom{x}}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$: x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$5) y = x - 3 \quad y = \cos^{-1}(x - 3), 2 <$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \frac{4 \ln(2x^2 - 3)}{2}, x > \frac{\sqrt{6}}{2}$$

$$12) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x <$$

Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$x^4 + C_1 y = \sqrt[3]{x}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$5) x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$9) y = x - 3$$

$$y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \frac{\ln(2x^2 - 3)}{2}, x > \frac{\sqrt{6}}{2}$$

$$12) \frac{e^{2y}}{\sqrt{62}} = x^3 - \frac{1}{2}$$

$$12) \frac{y^3}{3} = x + \frac{x^3}{3}$$

$$= \sqrt[3]{x^3 + 3x}, x < 0$$



Find the general solution of each differential equation.

$$1) \frac{dy}{dx} = \frac{x^3}{y^2}$$

$$2) \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$3) \frac{dy}{dx} = 3e^{x-y}$$

$$4) \frac{dy}{dx} = \frac{2x}{e^{2y}}$$

$$5) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$$

$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$12) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$$

$$1 + C_1 y = \sqrt[3]{x}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{2} = x^2 + C_1$$

$$y = \frac{\ln(2x^2 + C)}{2}$$

$$5) x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$y = \sqrt[3]{3e^x + 1}$$

$$9) y = x - 3$$

$$y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}, x > \frac{\sqrt{6}}{2}$$

$$12) \frac{e^{2y}}{2} = x^3 - \frac{1}{2}$$

$$x + \frac{x^3}{3}y = \sqrt[3]{x^3 + 3}$$

Find the general solution of each differential equation.

1)  $\frac{dy}{dx} = \frac{x^3}{y^2}$

2)  $\frac{dy}{dx} = \frac{1}{\sec^2 y}$

3)  $\frac{dy}{dx} = 3e^{x-y}$

4)  $\frac{dy}{dx} = \frac{2x}{e^{2y}}$

5)  $\frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \sqrt[3]{13}$

6)  $\frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln \frac{3e^3 + 2}{e^3}$

7)  $\frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = 0$

8)  $\frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$

9)  $\frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$

10)  $\frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = \frac{\ln 5}{2}$

11)  $\frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$

12)  $\frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = -\sqrt[3]{4}$

$$1 + C_1 y = \sqrt[3]{x}$$

$$2) \tan y = x + C$$

$$y = \tan^{-1}(x + C)$$

$$3) e^y = 3e^x + C$$

$$y = \ln(3e^x + C)$$

$$4) \frac{e^{2y}}{3} = x^2 + C$$

$$x^2 + C_1 y =$$

$$x^2 + \frac{1}{3}y = \sqrt[3]{3}$$

$$6) e^y = 2e^x + 3$$

$$y = \ln(2e^x + 3)$$

$$7) \tan y = x - 3$$

$$y = \tan^{-1}(x - 3)$$

$$8) \frac{y^3}{3} = e^x + \frac{1}{3}$$

$$= \sqrt[3]{3e^x + 1}$$

$$9) y = x - 3$$

$$y = \cos^{-1}(x - 3), 2 < x < 4$$

$$11) \frac{e^{2y}}{2} = x^3 - \frac{1}{2} \frac{4 \ln(2x^2 - 3)}{2}, x > \frac{\sqrt{6}}{2}$$

$$12) \frac{e^{2y}}{\sqrt{62}} = x^3 - \frac{1}{2}$$

$$x + \frac{x^3}{3}y = \sqrt[3]{x^3 + 3}$$